DG Interns Hub – Cybersecurity Internship

Week 1 Assignment Report: Networking Fundamentals

Author: B Sai Amulya Goud

Intern Id: DG/SEPTEMBER/CYBER/179

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Program Details:

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TASK 1: OSI MODEL

- OSI MODEL (OPEN SYSTEMS INTERCONNECTION) HAS 7 LAYERS is a conceptual framework developed by the International Organization for Standardization (ISO) to explain how communication takes place between different devices over a network.
- The OSI model divides the network into 7 different layers. These are made for better network understanding, troubleshooting etc.
- Each layer has a specific function to do.
- The 7 different layers of OSI models are:

Layer 7: Application Layer

- The Application Layer is the seventh and highest layer of the OSI model. It provides network services directly to applications that users interact with.
- It allows software applications to communicate over a network.
- Ex: HTTP, HTTPS, FTP

Layer 6: Presentation Layer

- The Presentation Layer is responsible for making sure that data sent by one system can be correctly understood by another system.

- The main functions of it are data encryption and decryption, data compression
- Ex: Turns readable data into encrypted version and transmits the data.

Layer 5: Session Layer

- The Session Layer establishes, manages, and terminates communication sessions between applications.
- A session is a period of communication between two systems.
- The main functions of it are to establish a session, maintain the session, synchronize the communication, terminate the session.
- Ex: Video Conferencing session within 2 users. The communication needs to be established and maintained through the end.

Layer 4: Transport Layer

- The Transport Layer provides end-to-end communication between devices.
- The main functions of it are segment the data, data delivery, error detection and data recovery,

flow control and managing communication between applications

- It has protocols those are TCP (Transmission Control Panel) and UDP (User Diagram Protocol)
- Ex: When downloading a PDF from a website, TCP helps ensure that all the pieces of the file arrive correctly.

Layer 3: Network Layer

- The Network Layer is responsible for delivering packets between different networks.
- Its major functions include Logical addressing, routing, packet forwarding, path selection.
- The most important protocol associated with this layer is Internet Protocol.
- Ex: sending the message from one country to another. The message is checked in various servers.

Layer 2 : Data Link Layer

- The Data Link Layer provides reliable communication between devices that are directly connected within the same network segment.

- Its main function is to framing, MAC addressing, error detection, controlling access to the transmission medium.
- Ex: Computer connected to the ethernet cable and computer sends ethernet frame to the switch. It examines the frame and sends it to the right device

Layer 1: Physical Layer

- The Physical Layer is the lowest layer of the OSI model. It deals with the actual transmission of raw bits 0s and 1s through a physical or wireless medium.
- Its characteristics are cables, connectors and electric signals.
- Ex: Ethernet cables, Fiber-optic cables, Network connectors

TASK 2: TCP/IP

- The TCP/IP (Transmission Control Protocol/Internet Protocol) Model is a networking framework used to describe how devices communicate over the Internet and other computer networks.

- the TCP/IP model commonly has 4 layers. They are

Layer 4: Application Layer

- The Application Layer is the highest layer of the TCP/IP model. It provides network services to applications that users interact with.
- It combines the functions of the Application, Presentation, and Session layers of the OSI model.
- Its main functions are Providing network services to applications, Data representation and formatting, Encryption-related functions, Supporting different network protocols
- Its common protocols are HTTP, HTTPS, FTP
- Ex: when we type browser name on the search engine, the application layers use protocols like DNS and HTTPS to connect to google servers.

Layer 3: Transport Layer

- The Transport Layer provides communication between applications running on different devices.
- It corresponds to the Transport Layer of the OSI model.

- Its main functions are end to end communication, Data segmentation, reliable delivery, error control and flow control.
- Its main protocols are TCP and UDP
- Ex: When you download a document from a website, TCP helps ensure that the complete document reaches your computer correctly.

Layer 2: Internet Layer

- The Internet Layer is responsible for delivering packets between different networks. It corresponds approximately to the Network Layer of the OSI model.
- Its main functions are logical addressing, packet routing, packet forwarding
- The important protocols are IP, ICMP, ARP
- Ex: When you send data from your computer to a website server, routers use destination IP addresses to determine where packets should be forwarded.

Layer 1: Network Access Layer

- The Network Access Layer is the lowest layer of the TCP/IP model.

- Its main functions are Physical transmission of data, Frames, MAC addresses and Network hardware
- Ex: Ethernet, Wi-Fi etc

Difference between OSI model and TCP/IP model

OSI MODEL	TCP/IP MODEL
Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
The number of layers are 7	The number of layers are 4
It is developed by ISO	It is developed by DARPA/DoD research community
It is mainly used for learning, design and troubleshooting	It is main used for widely used in real-world Internet networking
It has separate layers	It has internals layers
The Model is protocol-independent	The model built around protocols such as TCP/I

TASK 3: NETWORK PROTOCOLS

- A network protocol is a set of rules that defines how devices communicate and exchange data over a network.
- HTTP / HTTPS: Governs communication between web browsers and web servers
- Ex: When you visit a website such as a news website, your browser requests web pages, images, and other resources from the server.
- FTP: File Transfer Protocol is a protocol designed for transferring files between computers over a network.
- Ex: Suppose a web developer needs to upload website files to a hosting server. FTP can be used to transfer those files.
- DNS: Domain Name System is a system that translates human-readable domain names into IP addresses.
- Ex: when a website name is typed the device needs to determine the IP address associated with that domain.

- DHCP: Dynamic Host Configuration Protocol automatically provides network configuration information to devices joining a network.
- Ex: When you connect your smartphone or laptop to your home Wi-Fi, you normally don't manually enter an IP address.

TASK 4: IP ADDRESSING

- IP Addressing is a logical address assigned to a device connected to a network. It is used to identify the device and help deliver data to the correct destination.
- Types of IP Address
- IPv4: IPv4 is the most commonly encountered IP addressing format. An IPv4 address consists of 32 bits, divided into four groups called octets.
- IPV6: IPv6 was developed to provide a much larger number of addresses than IPv4. An IPv6 address consists of 128 bits.

Public vs Private IP Address

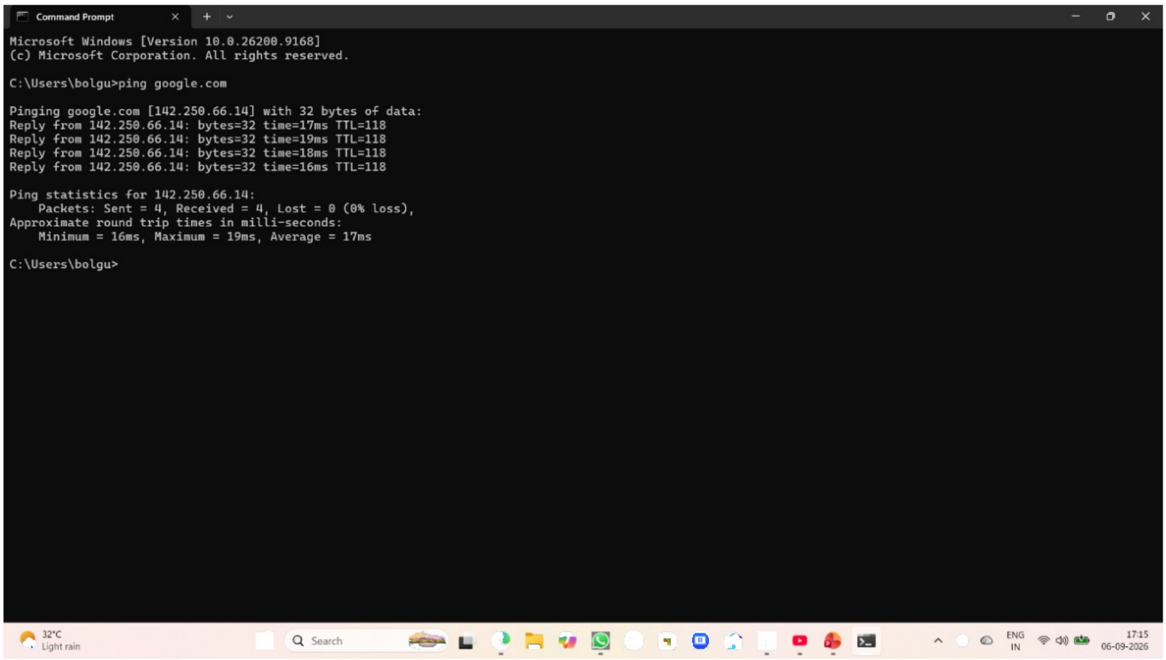
- A public IP address is an IP address that is used to identify a network or device on the public Internet.
- Public IP addresses need to be globally unique within the Internet's addressing system.
- A private IP address is used inside a private network, such as a home, school, office, or organization.
- Ex: ISP-assigned address for public IP address and 192.168.1.10 is for private address

Basic subnetting concept

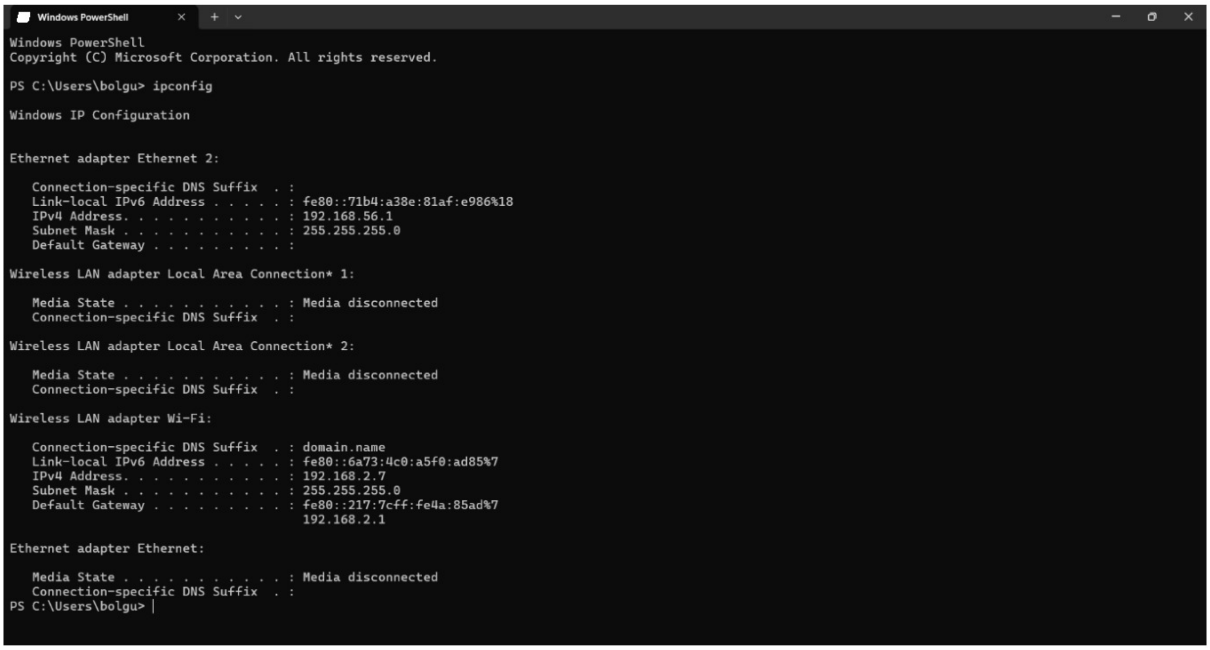
- Subnetting is the process of dividing a larger IP network into smaller logical networks called subnets.
- Subnetting helps organizations use IP addresses efficiently and divide networks into manageable sections.
- A subnet mask is used with an IPv4 address to determine which portion represents the network and which portion represents the host.
- Ex: normal network: 192.168.1.0/24 Subnet mask: 255.255.255.0

TASK 5: NETWORK COMMANDS

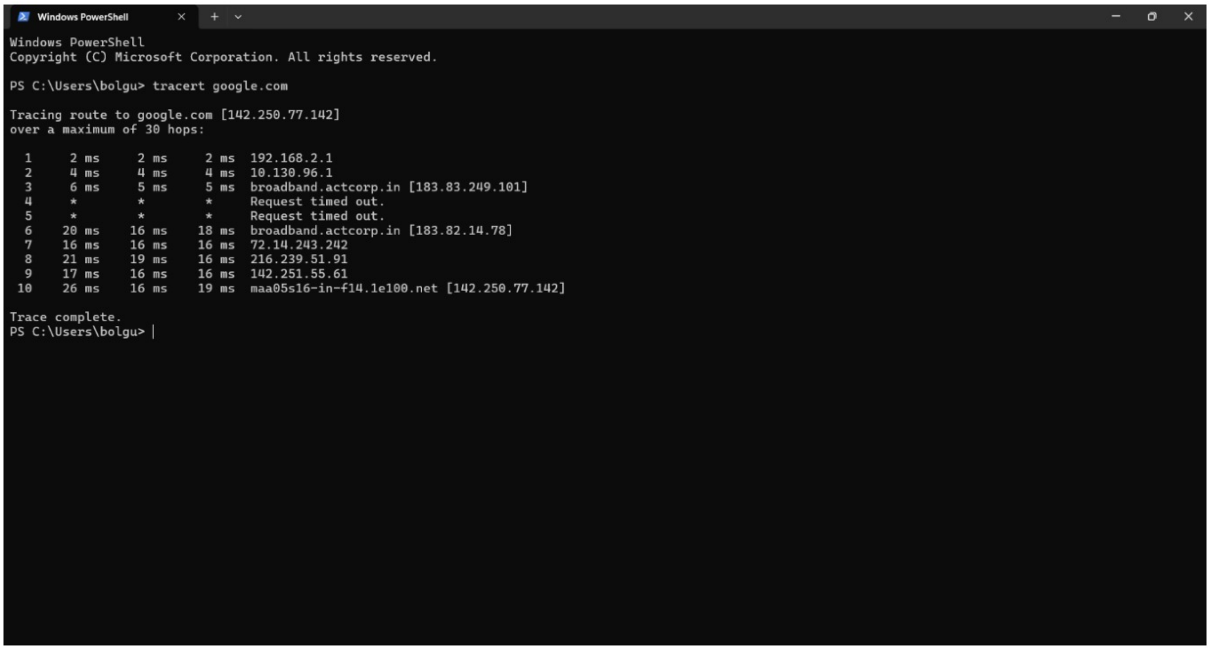
- Ping google.com



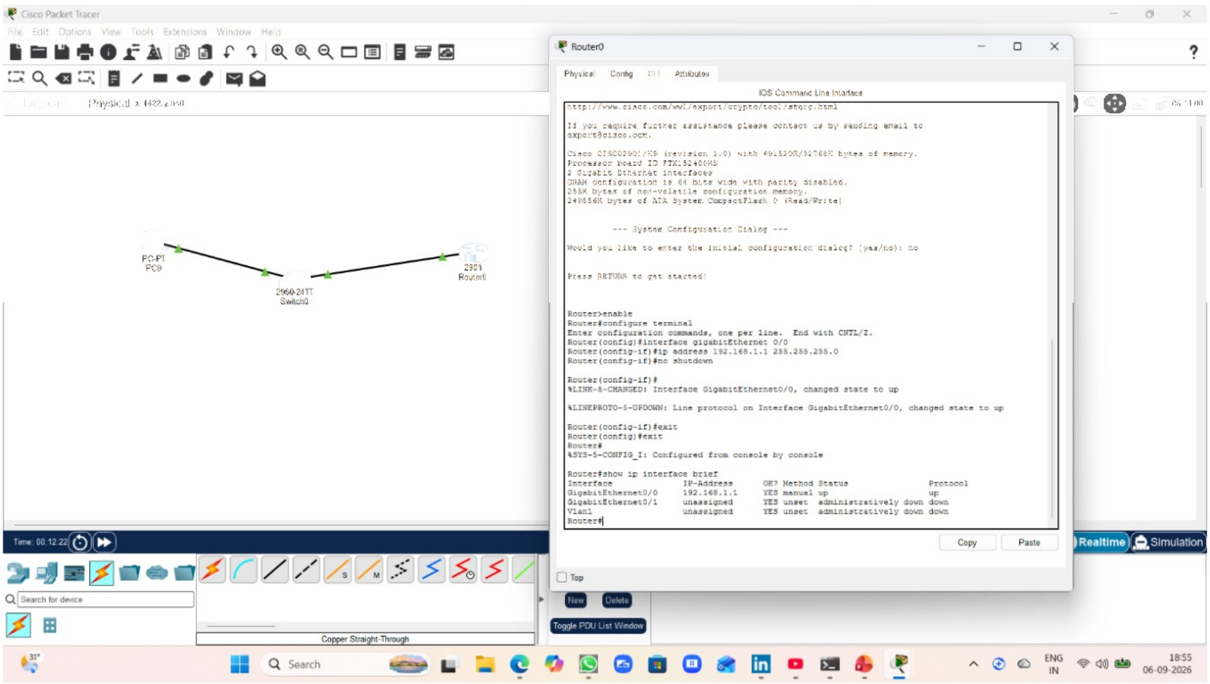
- IP Configuration



- Tracret google.com



TASK 6: NETWORK SIMULATIONS



- Ping 192.168.1.1